

Lab 04: Preferences, Priors, and Precision Tuning

Objective

Explore how C, D, E vectors and precision γ shape agent behavior.

Prerequisites

- Completed Labs 01–03
- Understanding of EFE decomposition (risk + ambiguity)

Part 1: C-Vector Effects

Goal: Compare agent behavior with different preference vectors.

1. Create a T-maze `GenerativeModel` (4 states, 3 obs, 3 actions).
2. Run the perception-action loop for 15 steps with three C-vectors:
3. `C_neutral = [0, 0, 0]` (no preferences)
4. `C_reward = [0, 3, -3]` (prefer reward)
5. `C_explore = [0, 0, 0]` with a noisy A-matrix (epistemic drive)
6. Record and compare the action sequences and states visited.

```
# TODO: Create models with different C-vectors
# TODO: Run loops and compare
```

Response:

Write your response here

Part 2: D-Vector Effects

Goal: Investigate how the initial state prior affects early behavior.

1. Fix C and E. Create two agents with:
2. `D_certain = [1, 0, 0, 0]` (confident at center)
3. `D_uncertain = [0.25, 0.25, 0.25, 0.25]` (uniform)
4. Run 5 steps. Compare the first action selected and the VFE trajectory.
5. Visualize both D-vectors using `plot_D_prior()`.

Response:

Write your response here

Part 3: E-Vector and Habit Formation

Goal: Demonstrate how habits bias action selection.

1. Create an agent with `E = [0.8, 0.1, 0.1]` (strong stay habit) and $\gamma = 0.5$.
2. Create an agent with `E = None` and the same γ .
3. Run both for 20 steps. Count how often each agent selects action 0 (stay).
4. Visualize habits using `plot_E_habits()`.

Response:

Write your response here

Part 4: Precision Sweep

Goal: Map the exploration–exploitation tradeoff across γ values.

1. For $\gamma \in \{0.1, 0.5, 1.0, 2.0, 4.0, 8.0, 16.0\}$:
2. Run `run_policy_inference()` with the same beliefs and C-vector
3. Record the resulting $q(\pi)$
4. Visualize with `plot_precision_sweep()`.
5. Identify the γ value where the agent transitions from near-random to near-deterministic.

```
from active_inference.math import run_policy_inference
from active_inference.visualization import plot_precision_sweep

# TODO: Sweep  $\gamma$  and collect  $q(\pi)$  matrix
```

Response:

Write your response here

Part 5: Analysis Questions

1. Did the agent with `C = [0, 3, -3]` consistently reach the reward arm? If not, what prevented it?
2. How many steps did the habit-biased agent waste on action 0 compared to a habit-free agent?
3. At which γ value did policy selection become effectively deterministic? How does this relate to the magnitude of EFE differences?

Response:

Write your response here

Summary

Skill	Library Component	Status
Design C-vectors for goal-directed behavior	<code>GenerativeModel(C=...)</code>	
Visualize preferences, priors, and habits	<code>plot_C_preferences()</code> , <code>plot_D_prior()</code> , <code>plot_E_habits()</code>	
Investigate precision effects	<code>run_policy_inference(gamma=...)</code>	
Map the exploration–exploitation tradeoff	<code>plot_precision_sweep()</code>	
Compare agents with different cognitive configurations	Multi-agent experiment design	